

Theory Lectures on

Manufacturing Processes (MEI102)

CHM – Chemical Machining

ECM – Electro-Chemical Machining

(Lecture 6)



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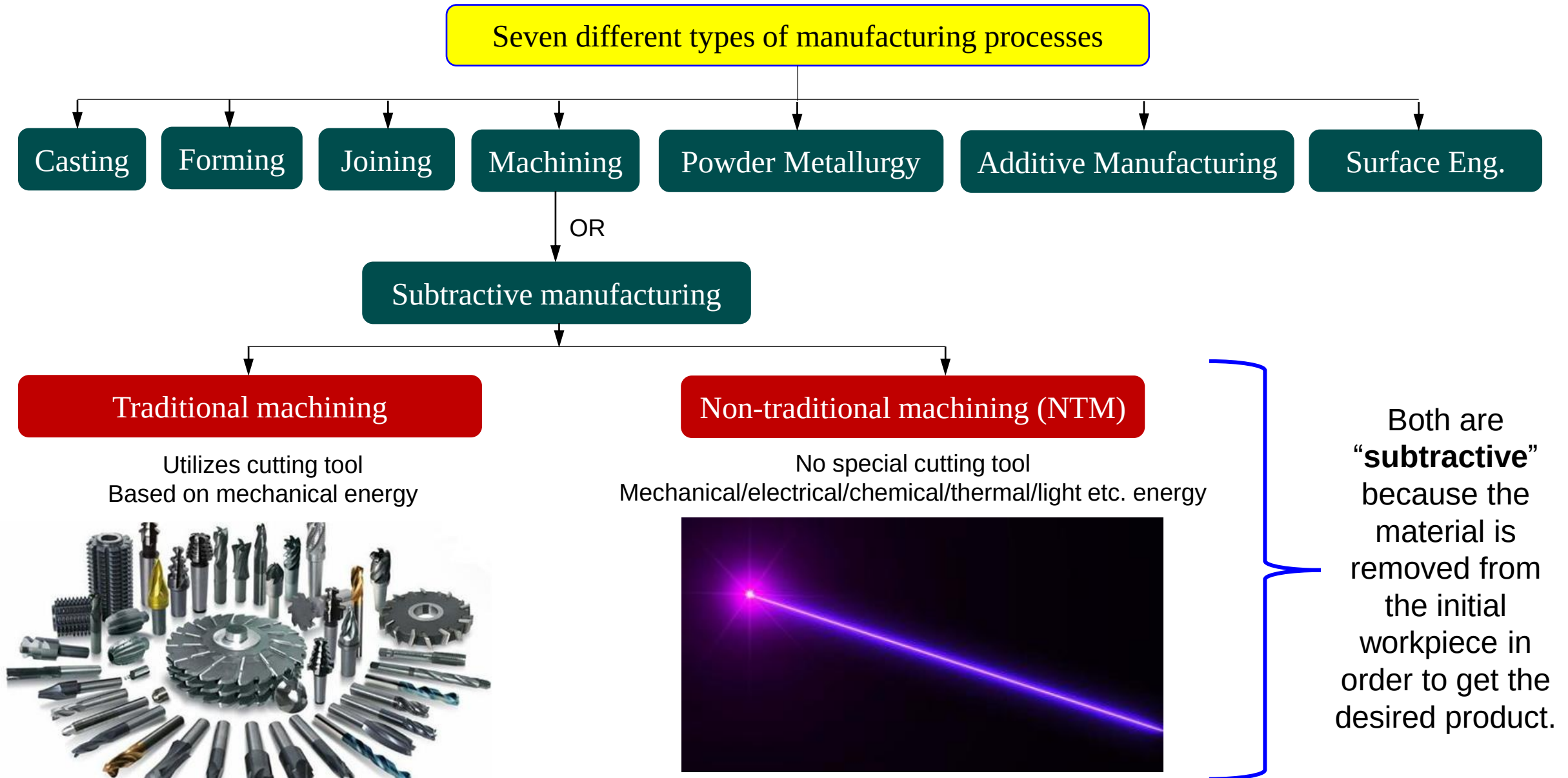
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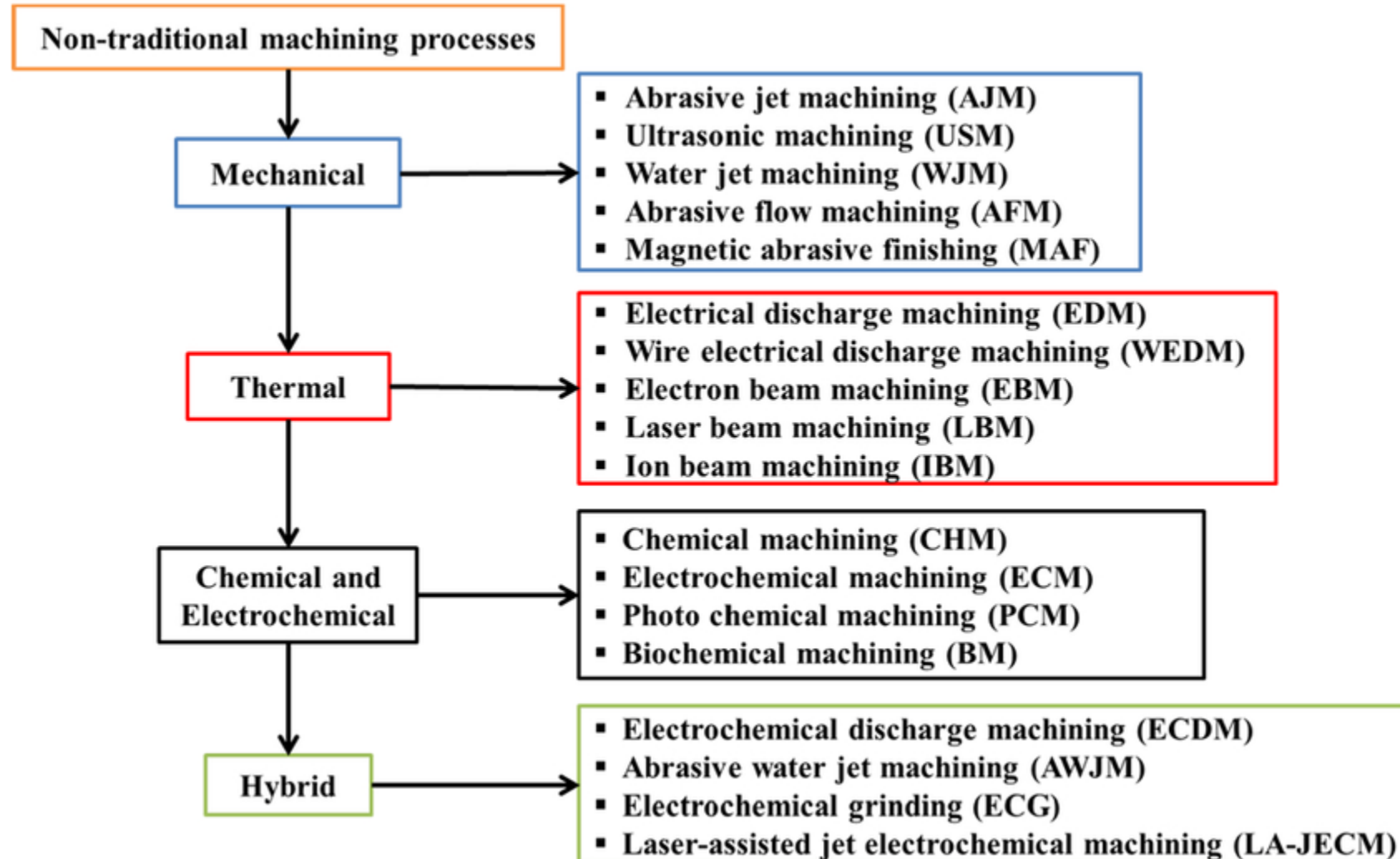
Department of Mechanical Engineering



Traditional and Non-Traditional Machining

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Chemical Machining (CHM)



Chemical machining is nothing but controlled dissolution of the workpiece material through a chemical reaction induced by a strong chemical reagent.

Workpiece Material + Chemical reagent $\Rightarrow$ Metal halides + Other biproducts

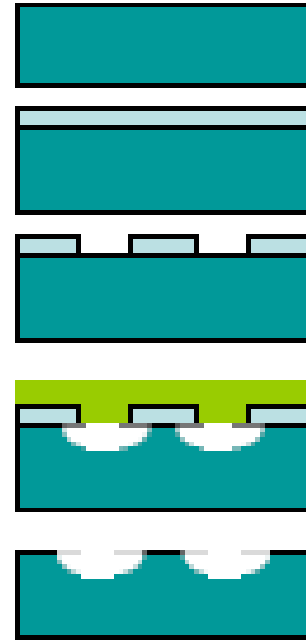
Reagent can be either Acidic or Alkaline.





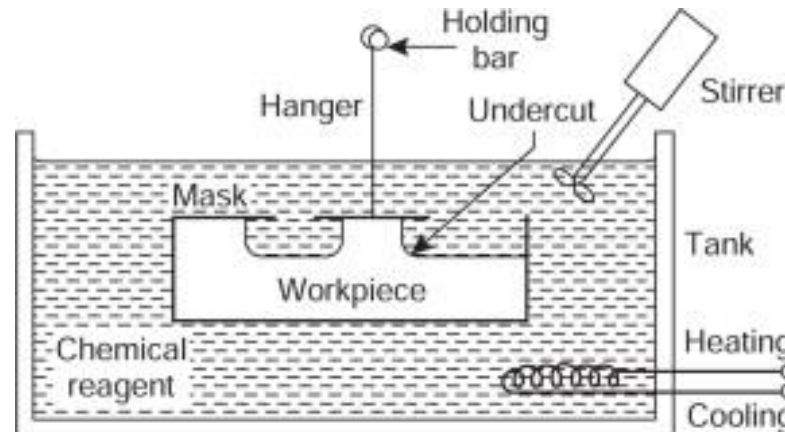
Working principle of CHM

- Working principle of chemical machining is based on selective chemical dissolution and subsequent removal of material (etching) from the workpiece using reactive acidic or alkaline solution.
- The part of the workpiece metal from where material is to be removed is brought into contact with a strong corrosive chemical called etchant.
- The etchant chemically reacts with the workpiece causing slow dissolution of the solid material. Thus the metal is removed by the chemical attack of the etchant.
- Mechanism of material removal => **Chemical dissolution.**
- Chemically resistant coatings (or masks) are used to protect the surfaces that are not to be machined.

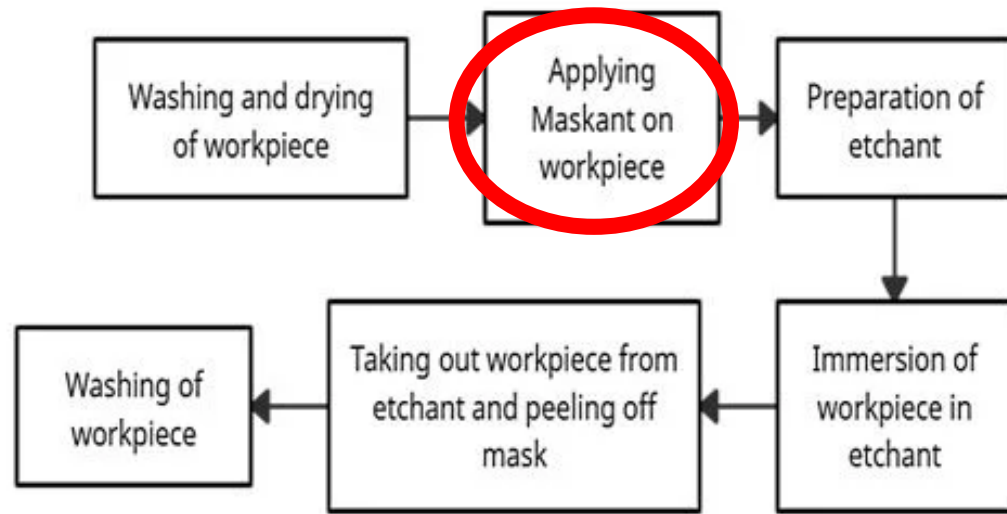


General processing steps of CHM

- Clean the workpiece surface - to remove dirt, oil, coating, etc. to insure uniform etching
- Apply maskant - a maskant (neoprene, polyvinylchloride, polyethylene, and other polymers) is chemically resistant to etchant and is deposited on the portions of work surface that need not be etched
- Selectively remove maskant – if required, cut and peel the maskant to expose the workpiece surface that need to be etched.
- Etch - Part is immersed in etchant which chemically attacks those exposed portions. Control the etching time to get desired depth of cut.
- Remove maskant and clean.



- Stirring helps supply of fresh etchant at the exposed work surface.
- Controlling temperature helps accelerating/controlling the chemical reaction rate.



Workpiece material	Maskant material
Aluminum and its alloys	Butyl rubber, Polymer, and Neoprene
Copper and its alloys	Polymer
Iron-based alloys	Butyl rubber and Polymer
Nickel	Neoprene
Magnesium	Polymer
Titanium	Polymer

Cut and Peel Masking Method

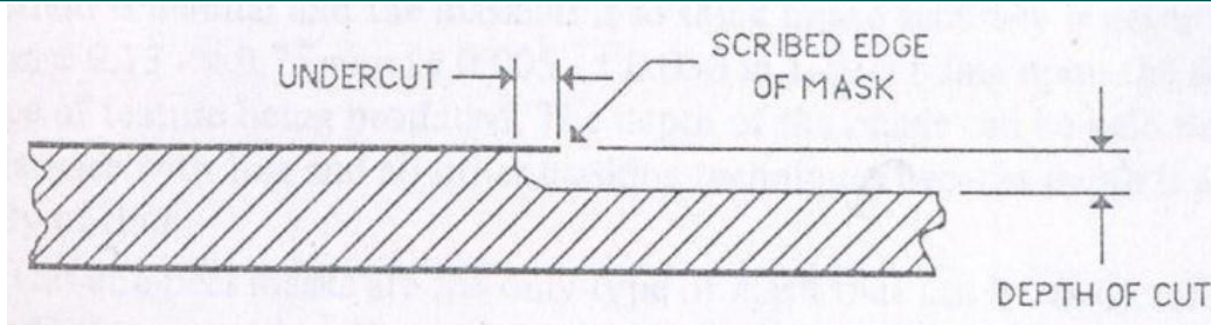
- Maskant is applied on the part by dipping, painting, or spraying
- After maskant hardens, it is cut by hand using a scribing knife and peeled away in areas of work surface to be etched
- Typical Accuracy: $\pm 0.13 - 0.75$ mm
- Used for large workpieces, deep etch depths (1.5 mm), low production quantities, and where accuracy is not a critical

Screen Resist Method

- Maskant applied by silk screening methods
- Maskant is painted through a silk or stainless steel mesh containing stencil onto surface areas that are not to be etched
- Typical Accuracy: $\pm 0.05 - 0.18$ mm
- Used for mainly in fabrication of printed circuit boards (PCB)

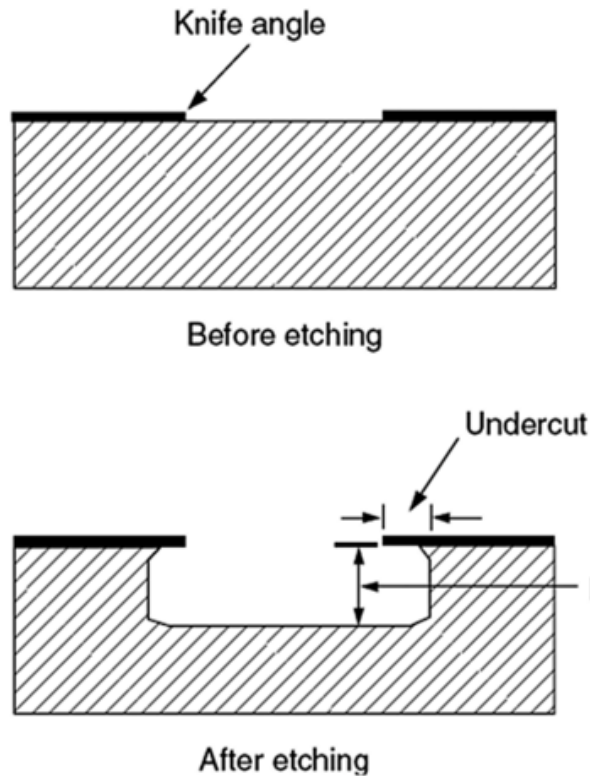
Photographic Resist Method

- Masking materials contain photosensitive chemicals (photoresist is a polymer that adheres to the metal when exposed to UV light).
- Maskant is applied to work surface and exposed to light through a negative image of areas to be etched
- Typical Accuracy: ± 0.025 mm
- Used for mainly for integrated circuits and PCB fabrication



Etching downward => Depth of cut (desirable)

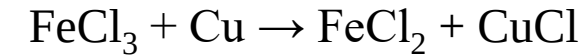
Etching sideways under maskant => Undercut (undesirable)



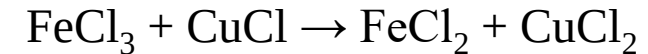
Factors for selecting etchant

- Work material
- Depth and rate of material removal
- Surface finish requirements and
- Compatible with the type of maskant

For example, say the workpiece material is copper. This work material can be etched by ferric chloride (FeCl_3).



Thus, copper reacts with ferric ions and forms cuprous chloride (CuCl) and ferrous chloride (FeCl_2). The CuCl further reacts in the etchant solution to produce CuCl_2 as follows:



The built-up CuCl_2 again reacts with copper and forms CuCl :



Workpiece material	Etchant
Aluminum and its alloys	FeCl_3
Steel	FeCl_3
Brass or bronze	NH_4HF_2
Copper and its alloys	FeCl_3 or CuCl_2
Magnesium and its alloy	H_2SO_4
Nickel	FeCl_3
Titanium	$\text{HF} + \text{HNO}_3$



Advantages of CHM

- Easy weight reduction
- No effect of workpiece materials properties such as hardness
- Simultaneous material removal feasible from multiple portions
- No burr formation. So no protrusion at the edges
- No stress introduction to the workpiece
- Low capital cost of equipment
- Easy and quick design changes
- Requirement of less skilled worker
- Low tooling costs
- The good surface quality

Disadvantages of CHM

- Difficult to get sharp corners due to inherent undercut
- Difficult to chemically machine thick material (limit is depended on workpiece material, but the thickness should be around 10 mm. Otherwise excessive undercut can occur)
- Scribing accuracy is very limited,
- Most of the chemicals such as cleaning solutions, etchants, strippers etc. are very hazardous liquids. Therefore, handling is costly and disposal creates environmental issues.
- Etchants are also very dangerous for workers

Applications of CHM

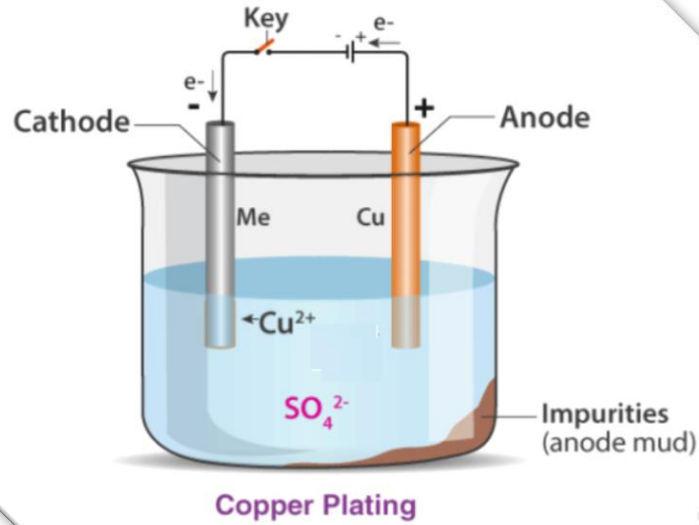
- Uniform weight reduction from bulk metallic structures (by removing a thin layer of material from entire exposed surfaces – this is very useful for aircraft bodies)
- Removal of altered or stressed layer from conventionally machined components.
- Ideal for micro-detailing on workpieces.
- Precision machining of intricate parts, especially on brittle and hard materials such as glass, ceramics, etc.
- Engraving of designs with high precision.
- Fine finishing and deburring of workpieces.
- Crack-free component manufacturing.

Electro-Chemical Machining (ECM)



Do you remember electroplating or electro-galvanizing?

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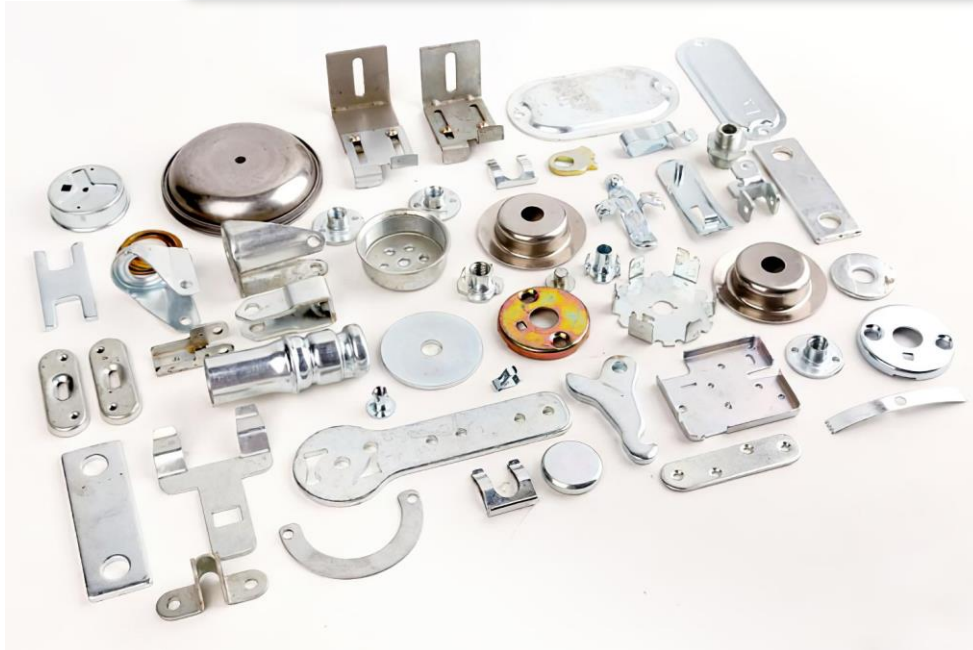


Electroplating

- Electroplating is the process of coating an object by a layer of metal using electro-chemical dissolution principle.
- Anode (positive electrode) is made of the same material that needs to be deposited. Anode material itself undergoes dissolution and deposits over the cathode.
- Cathode (negative electrode) is the workpiece material on which a coating layer will be deposited.
- Electrolyte is a solution of a salt of the metal to be coated.

Now think the reverse scenario

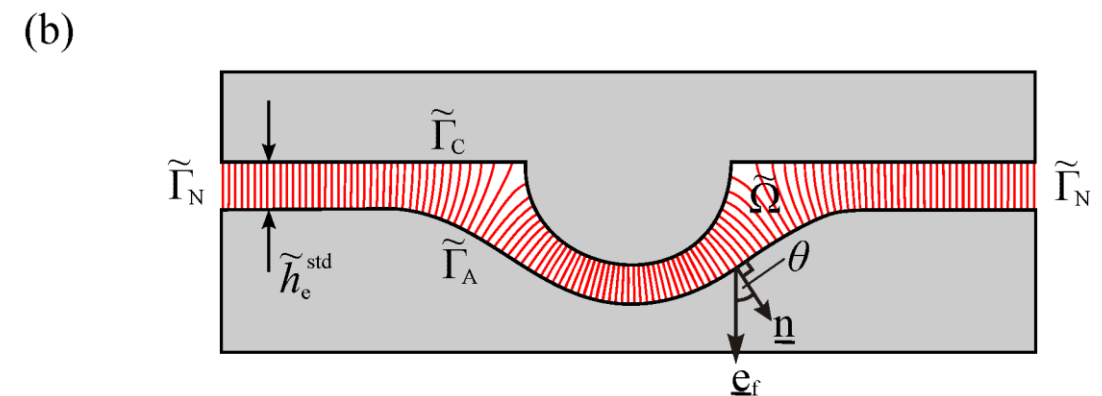
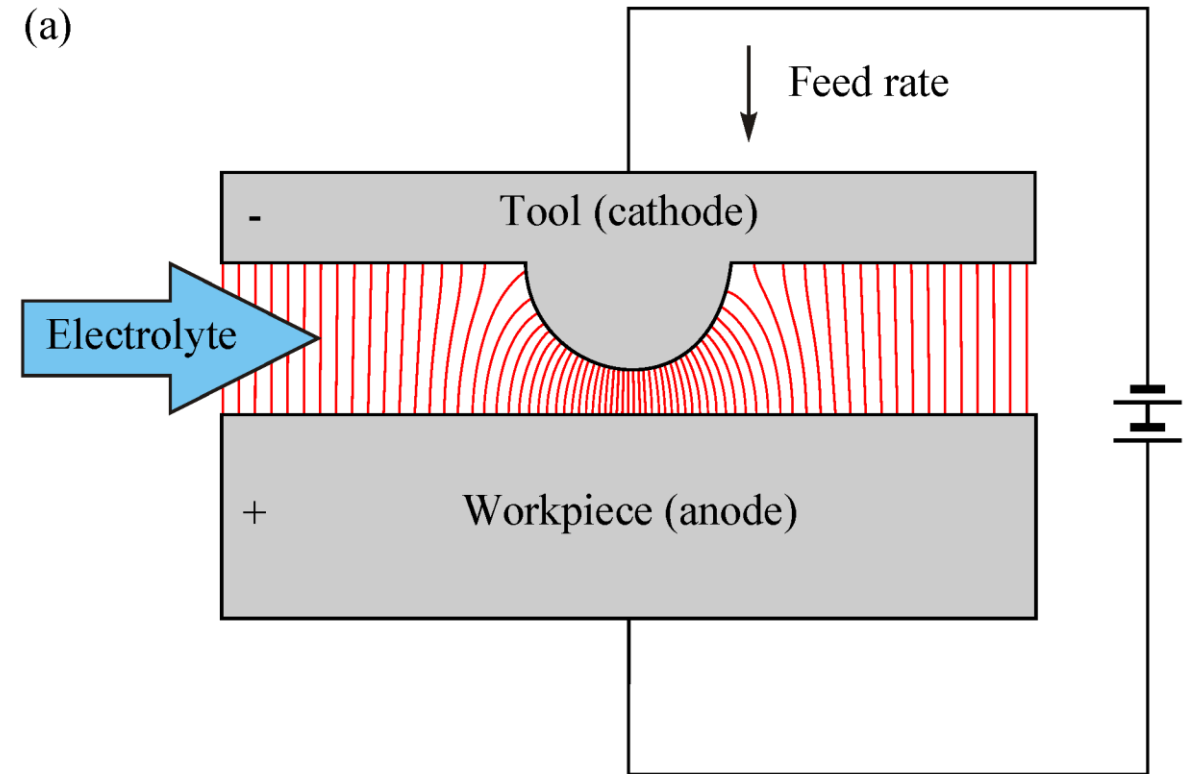
Instead of depositing a layer of coating on the workpiece, can a thin layer of material be removed from the workpiece?

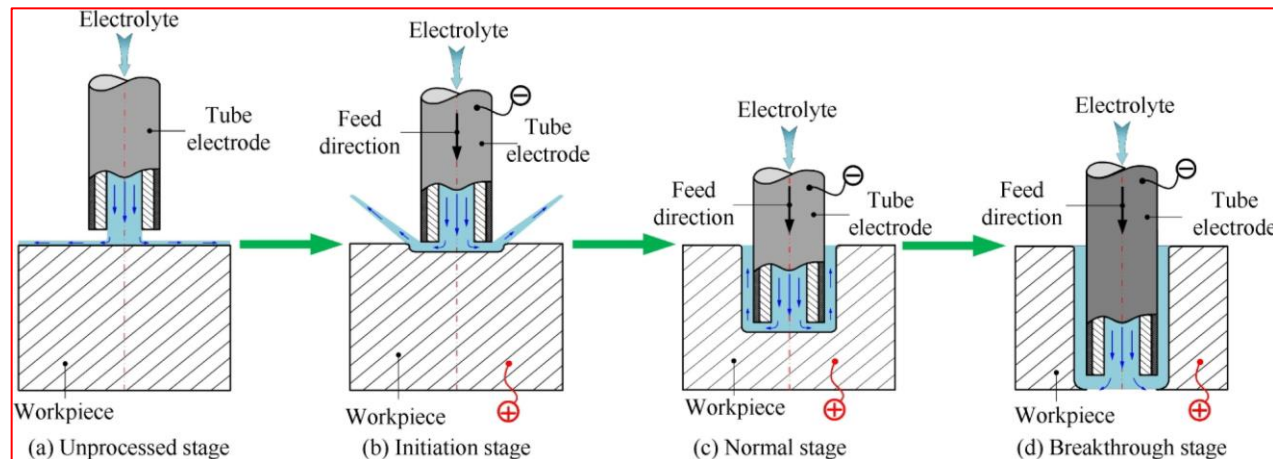
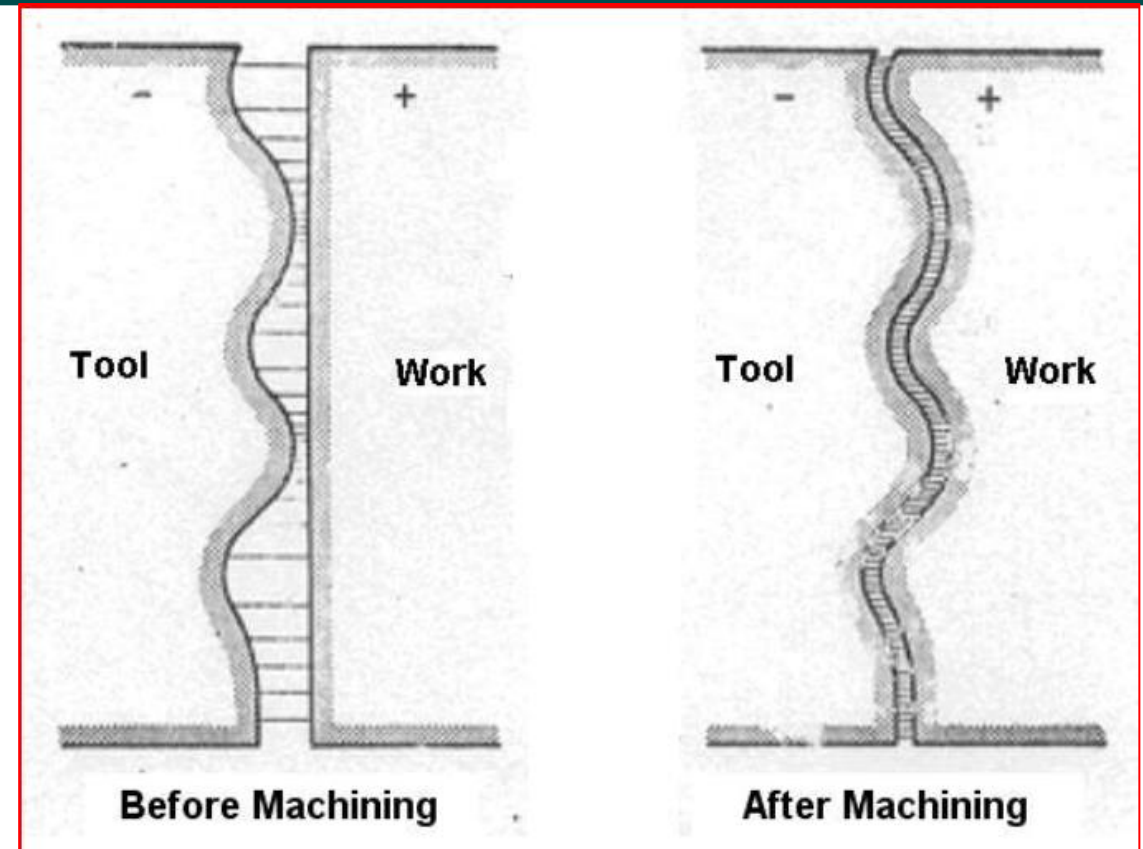
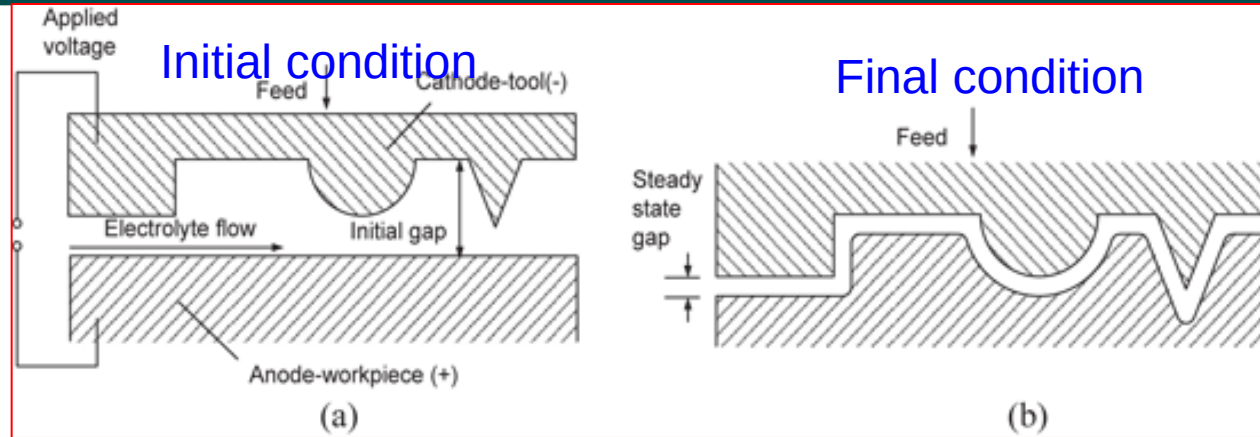


Electroplating	Electro-Chemical Machining
Workpiece => Cathode (negative)	Workpiece => Anode (positive)
Tool => Anode (positive)	Tool => Cathode (negative)
Anode/tool material dissolves and deposits on cathode/workpiece.	Anode/workpiece material dissolves but does not deposit on tool/cathode (remains in solution)
Coating occurs on the workpiece.	Removal occurs from workpiece.

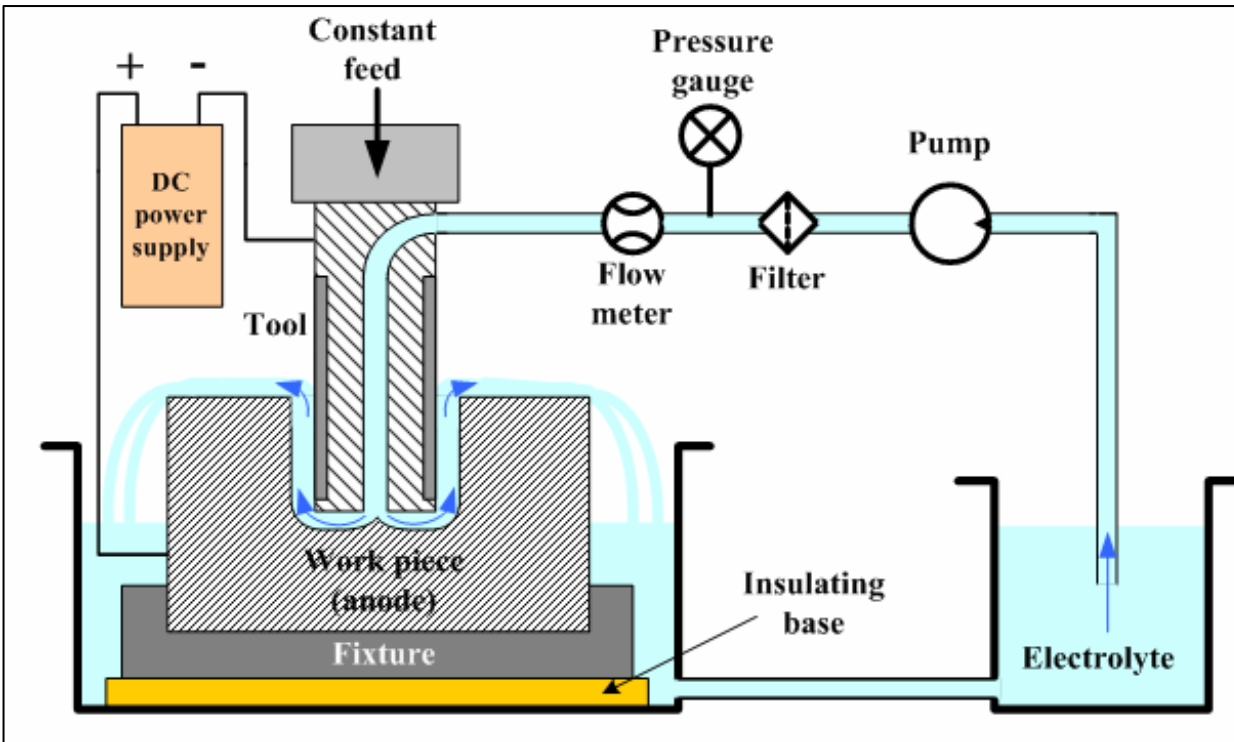
Working principle of ECM

- Electrochemical machining (ECM) is an unconventional machining process where material removal is accomplished through controlled anodic dissolution of workpiece.
- Workpiece is immersed in an electrolyte and connected to positive terminal of a low-voltage (5 – 20 V), high-current (1000 A) DC power supply. Workpiece acts as an electrode.
- The tool (another electrode) is also immersed at some distance from the workpiece and connected to the negative terminal. The tool is also given an inverse profile to the desired shape.
- The gap between the tool-electrode and workpiece-electrode is called Inter-Electrode Gap (IEG). Fresh electrolyte is continuously supplied at this IEG.
- The tool is slowly advanced towards the stationary workpiece causing electro-chemical reaction to take place. This results in gradual dissolution/removal of material from workpiece. The dissolved ions remain in the electrolyte.
- After considerable time of such electro-chemical reaction, the workpiece obtains the inverse profile of the tool.



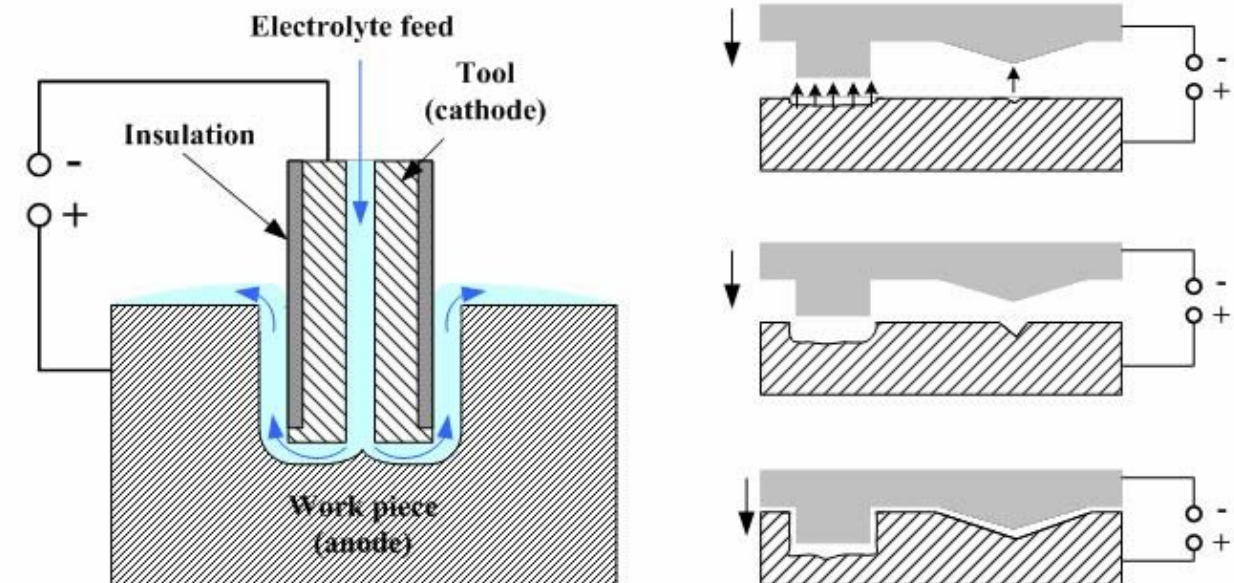


The tool profile must be inverse to the desired profile. If the desired profile is convex, the tool profile must be concave. This is because the electro-chemical reaction (ion flow) will be high at the closest points (least potential difference). Tool shape does not change during the process (tool remains intact; only workpiece material dissolves).



Inter-Electrode Gap (IEG) is very small $\Rightarrow 0.1 - 0.5$ mm

- Electrolyte must be continuously circulated to ensure fresh electrolyte supply at the Inter-Electrode Gap (IEG).
- Note that the dissolved ions (from workpiece) remains suspended in the electrolyte. If such dissolved ions are not removed from the IEG, then short circuit will occur that will prohibit further electro-chemical reaction between tool and workpiece.



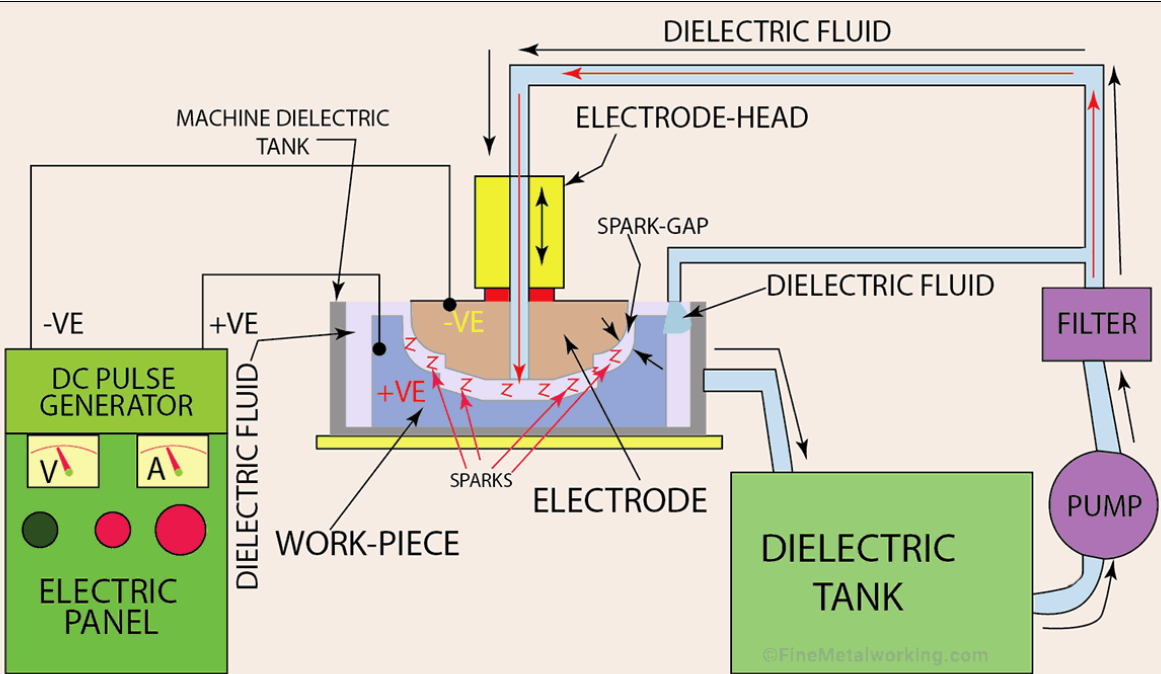


The electrolyte should possess several important properties:

- Electrolyte must possess high electrical conductivity.
- Electrolyte should possess high specific heat.
- The viscosity must be as low as possible.
- The electrolyte must be chemically stable.
- Electrolyte must be chemically active to cause the better metal or material removal rate.
- Electrolyte should not form any type of excess layer on the top of electrolyte, tool or the work piece.
- Electrolyte should not be toxic and corrosive.
- Electrolyte should be economical and easily available.

Workpiece material	Electrolyte
Ferrous alloys	Sodium chloride (NaCl), Sodium nitrate (NaNO ₃)
Nickle alloys	Hydrochloric acid (HCl)
Titanium alloys	HF + HCl + HNO ₃
Tungsten carbide	Sodium hydroxide (NaOH)
Magnesium and its alloy	H ₂ SO ₄
Nickel	FeCl ₃
Titanium	HF + HNO ₃

The products of the electrochemical reaction should be removed from the gap between the work piece and the tool. Accumulation of the reaction products causes decrease of the process efficiency and reduction of the rate of machining. Therefore the electrolyte flow speed should be high. Commonly it is in the range 300-3,000 m/min. The inlet pressure should be in between 0.15-3 MPa. The electrolyte system should comprise a fairly strong pump. The electrolyte is continuously filtered in order to trap the precipitated reaction products (sludge).



EDM WORKING PRINCIPLE

Dielectric is used in electric-discharge machining (EDM) process. This dielectric is electrically non-conductor in stable situation but when sufficient voltage is applied it undergoes “*dielectric breakdown*” resulting in a conductive channel formation between the electrode and workpiece for a very short period of time. Such conductive channel allows free flow of huge electrons within a short period resulting in the spark formation.

Dielectric	Electrolyte
Dielectric fluid is utilized in electric-discharge machining (EDM) process.	Electrolyte is utilized in electro-chemical machining (ECM) process.
A dielectric should be electrically non-conductor at normal condition; however, it should behave as good conductor after <i>dielectric breakdown</i> .	Electrolyte should always be good conductor of electricity for completing the electrical circuit to allow smooth flow of electrons and ions between two electrodes.
The dielectric should be oxygen free. Due to extreme spark heat, liberation of oxygen can oxidize the work surface.	Presence of oxygen in electrolyte does not matter as ECM process does not generate extreme heat.
Common dielectric fluids used in EDM include deionized water, transformer oils, kerosene, paraffin, etc.	Commonly used electrolytes in ECM process include sodium chloride (NaCl) and sodium nitrate (NaNO3) solutions.



Difference between EDM and ECM processes

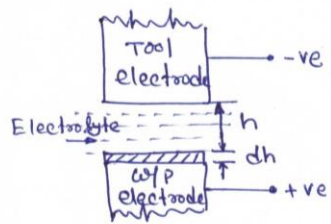
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Electro-Discharge Machining (EDM)	Electro-Chemical Machining (ECM)
EDM is one <i>thermal energy</i> based non-traditional machining process.	ECM is <i>electro-chemical energy</i> based non-traditional machining process.
Material removal takes place by <i>melting and vaporization (spark erosion)</i> . It is one volume removal process.	Material is removed by controlled <i>electro-chemical dissolution</i> of work metal (anode). Material removal takes place <i>in the form of ions of work metal</i> .
Owing to electric spark, significant <i>heat is generated</i> in EDM process. The localized temperature reaches as high as <i>10,000°C</i> .	<i>No heat generation</i> takes place in ECM. Accordingly, temperature at the machining zone does not increase.
An appropriate <i>dielectric fluid</i> (such as kerosene, paraffin, deionized water) is used in EDM to assist spark generation.	A suitable <i>electrolyte</i> (such as NaCl and NaNO ₃) is applied in ECM to assist flow of ions between two electrodes.
During EDM operation, <i>electrode material is also removed gradually</i> along with workpiece material. However, electrode material removal rate is much lower than work material removal rate.	<i>Electrode does not suffer any wear</i> during ECM process. Same electrode can be used repeatedly for a longer duration. Accordingly, electrode life is higher.
EDM process typically requires <i>high potential difference (50 – 400V) and moderate current (100 – 200A)</i> .	ECM process requires <i>moderate voltage (<50V) but very high current (>1000A)</i> .



Need to slowly feed the tool towards workpiece in ECM

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Consider an appropriate electrolyte is flowing through a parallel gap (h) between the tool and workpiece connected with a DC power supply of voltage V .

Now, if dh be the change in gap within the time dt owing to electrochemical reaction, then the ~~mass~~ mass of the material dissolved be

$$dw = \rho \cdot A \cdot dh \quad \text{--- (i) where } \rho = \text{density of w/p material}$$

$A = \text{c/s area of w/p material}$

Now apply Faraday's Law of Electrolysis:

First law: The mass of a substance deposited at any ~~sub~~ electrode is proportional to the amount of charge passed.

$$m \propto Q \quad \begin{array}{l} m = \text{mass of dissolved substance} \\ Q = \text{Electrical charge passed} = It \end{array}$$

Second law: The mass of a substance deposited on the electrode due to passing a certain amount of charge is proportional to its chemical equivalent weight.

$$m \propto E \quad E = \text{Equivalent weight} = \frac{\text{Atomic weight}}{\text{valency } (v)}$$

$$\therefore m \propto Q \cdot E \Rightarrow m = \frac{1}{F} Q \cdot E \Rightarrow m = \frac{1}{F} \left(\frac{M}{v} \right) (It) \quad \text{--- (ii)}$$

This is the expression for Material Removal Rate (MRR) in ECM.

Using this Faraday's law, the equation (i) can be written as

$$dw = \frac{1}{F} \left(\frac{M}{v} \right) I \cdot dt \quad \text{--- (iii)}$$

Equating eq (i) and (iii):

$$\rho A dh = \frac{1}{F} \left(\frac{M}{v} \right) I \cdot dt \Rightarrow \frac{dh}{dt} = \frac{1}{F} \left(\frac{M}{\rho \cdot v} \right) \left(\frac{I}{A} \right) \quad \text{--- (iv)}$$

Therefore, the rate of change of gap (dh/dt) is proportional to the current density

Again, if V is the applied voltage, R is the resistance of the electrolyte, and γ is the specific resistance of the electrolyte, then

$$I = \frac{V}{R} = \frac{V}{\left(\frac{\gamma h}{A} \right)} = \frac{V \cdot A}{\gamma \cdot h} \quad \text{--- (v) } \parallel \text{ Specific resistance } \Rightarrow \text{ Resistivity}$$

Combining eq (iv) and (v):

$$\frac{dh}{dt} = \frac{1}{F} \frac{M}{\rho v} \cdot \frac{V}{\gamma h}$$

For constant voltage:

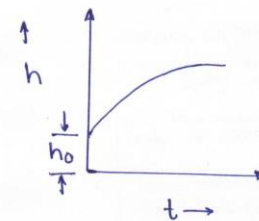
$$\frac{dh}{dt} = \frac{C}{h} \quad \text{--- (vi), where } C = \frac{1}{F} \frac{M}{\rho v} \frac{V}{\gamma} = \text{Constant}$$

Therefore, the rate of change of gap (dh/dt) is inversely proportional to gap (h).

Now say at time $t = 0$, the gap $h = h_0$

at time $t = t$, " " $h = h_1$

$$\text{on integration: } \int_{h_0}^{h_1} h \cdot dh = C \int_0^t dt \Rightarrow h_1^2 - h_0^2 = 2Ct \quad \text{--- (vii)}$$



Eq (vii) gives a parabolic variation of gap where the curve slope reduces to a negligible value at higher time (t). Therefore, the gap change rate or material dissolution rate falls with time. To maintain the material dissolution rate, the tool must be fed closer to the workpiece.

- Mathematically prove that the inter-electrode gap in ECM does not change appreciably after certain period of dissolution.
- Mathematically prove that the electrochemical dissolution rate in ECM approaches to zero after certain period of time.
- Mathematically demonstrate the need of slowly feeding the tool towards the workpiece to maintain the MRR constant.



Advantages of ECM

- ECM can machine highly complicated and curved surfaces in a single pass.
- A single tool can be used to machine a large number of pieces without any loss in its shape and size. Tool life is high
- Machinability of the work material is independent of its physical and mechanical properties.
- The process is capable of machining metals and alloys irrespective of their strength and hardness.
- Machining zone temperature does not increase, so there is no thermal damage on work surface.
- Machined surfaces are free from residual stress and burr
- It yields low scrap.

Disadvantages of ECM

- High capital cost of equipment
- Design and tooling system is complex
- Nonconductive material cannot be machined
- Corrosion and rust of ECM machine can be hazard

Applications of ECM

- ECM can be used to make disc for turbine rotor blades made up of HSTR alloys
- ECM can be used for copying of internal and external surfaces, cutting of curvilinear slots, machining of intricate patterns, production of long curved profiles, machining of gears and chain sprockets, production of integrally bladed nozzle for use in diesel locomotives, production of satellite rings and connecting rods, machining of thin large diameter diaphragms.